

Route G2-H: A Physical Source Budget and a Decisive Yield Bound

On-shell LHC Higgs production, not an invented dark beam

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Abstract

We select an existing accelerator process, Higgs production at the LHC, and the same conditional X -only portal as G2-X. Production and detection are tied to one coupling. A boost-independent, single-pass bound shows that even perfect interception of the Run-2 dark-particle budget cannot give a useful elastic xenon recoil sample in the retained energy window. This closes one physical realization without a detailed beam simulation. The direct Higgs source also erases the old sign preparation: opposite compact labels have identical timing, direction and momentum likelihoods. Neither result rules out the entire theory or constructs the original driven mode-conversion source.

1 Actual inputs and a different source preparation

Keep G2-X's tree EFT and mass matching:

$$\Delta\mathcal{L} = -\lambda_X H^\dagger H \sum_j |x_j|^2, \quad \mu_j = \Lambda \sqrt{25 + j^2}, \quad 0 \leq \lambda_X \leq \min(1, 50\Lambda^2/v^2). \quad (1)$$

The scale domain remains $1 \leq \Lambda \leq 30$ GeV. Use $m_h = 125.08$ GeV, $v = 246$ GeV, and $\Gamma_{\text{SM}} = .00410$ GeV as in G2-X. This branch uses the *drive-off* vacuum and ordinary Higgs decay, not a hidden implementation of $g(t)$. It neither supplies a prepared Φ beam nor performs the original ΦX conversion. The driven G2 population and its priors remain a separate, unrealized source hypothesis.

Choose one LHC interaction point's published 13 TeV Run-2 luminosity $\mathcal{L}_{\text{int}} = 139 \text{ fb}^{-1}$. The ATLAS production paper quotes the SM expectation $\sigma_h = 55.6 \pm 2.5 \text{ pb}$ and an agreeing measurement [1]. We use the *SM expectation*, consistent with the SM-production premise of the invisible-Higgs bound; the visible-channel measurement itself assumes SM branching fractions and is not an independent production input after changing those fractions. The small Higgs-mass convention difference in that production calculation is not resolved beyond its quoted precision.

$$N_h = \mathcal{L}_{\text{int}} \sigma_h = 7.7284 \cdot 10^6, \quad N_{X+\bar{X}} \leq 2N_h B_{\text{inv}}^{\text{max}} = 1.6538776 \cdot 10^6, \quad B_{\text{inv}}^{\text{max}} = .107. \quad (2)$$

The latter is the conditional observed ATLAS allowance at 95% CL [2], not a detected dark-particle sample. It assumes escaping invisible modes and SM production/visible widths. No combined confidence level is assigned to our subsequent theory/geometry bound.

The target column is based on XENON1T's approximately 2 tonne active cylinder, radius 47.9 cm, height 97 cm [3]. The detector is at Gran Sasso, not at the LHC; no installation, transport link or concurrent data collection is asserted. Perfect interception below is a deliberately generous mathematical envelope. Rounded geometry is sufficient to test an orders-of-magnitude shortfall.

2 The source spectrum and flux functional

For the diagonal vertex $-i\lambda_X v$, two-body phase space gives

$$\beta_j = \sqrt{[1 - 4\mu_j^2/m_h^2]_+}, \quad \Gamma_j = \frac{\lambda_X^2 v^2}{16\pi m_h} \beta_j, \quad (3)$$

$$B_j = \frac{\Gamma_j}{\Gamma_{\text{SM}} + \sum_k \Gamma_k}, \quad p_j^* = \frac{m_h \beta_j}{2}, \quad E_j^* = \frac{m_h}{2}. \quad (4)$$

Kinematic statements below concern open modes only. Each signed label is a distinct complex field and its decay creates one particle and one antiparticle. Their compact momenta are $j\Lambda$ and $-j\Lambda$; a uniform Higgs conserves their sum. Do not put another factor of two in Γ_j or reuse the old selected-collision weights. The twofold particle yield is already explicit in (2).

Let $dF_h(P)$ be a normalized physical Higgs four-momentum distribution, and let L_P boost its rest frame to the laboratory. The single-particle production measure is

$$\frac{dN_{X+\bar{X}}}{d^3p} = N_h \sum_j B_j \int dF_h(P) \int \frac{d\Omega_*}{4\pi} \sum_{\epsilon=\pm 1} \delta^3(\mathbf{p} - [L_P(E_j^*, \epsilon p_j^* \hat{\mathbf{n}}_*)]_{\text{spatial}}). \quad (5)$$

It integrates to $2N_h \sum_j B_j$. A real LHC source is boosted; setting every Higgs at rest is not a physical flux prediction. Published marginals alone are not silently promoted to a full correlated source, transport and timing sample.

For a point source at distance d , a surface element subtending $d\Omega = (\cos\theta/d^2)dA$ has fluence

$$\frac{d\mathcal{F}_j}{dE dA} = \frac{\cos\theta}{d^2} \frac{dN_j}{dE d\Omega} T_j(E, \Omega), \quad 0 \leq T_j \leq 1, \quad (6)$$

under straight, nonmultiplying transport. Fluence is cm^{-2} , not $\text{cm}^{-2}\text{s}^{-1}$. A flux additionally needs the actual luminosity time profile, travel time and detector live time. There is no reason to divide the Run-2 total by XENON's 180.7-day search interval or to multiply by its kg-day exposure again.

More generally an incident particle with momentum p on path γ contributes, in the dilute elastic approximation,

$$P_{\text{acc}} = \int_{\gamma} d\ell n_{\text{Xe}} \sum_A \xi_A \int_{E_0}^{E_1} dE_R \frac{d\sigma_A}{dE_R} \mathcal{R}_{\text{ROI}}(E_R, \mathbf{x}, t), \quad (E_0, E_1) = (.7, 50) \text{ keV}. \quad (7)$$

The released S2 matrix supplies a *volume-averaged* response, not an arbitrary narrow beam's spatial response. Our decisive bound uses $\mathcal{R}_{\text{ROI}} \leq 1$ everywhere, so it does not rely on applying that average where invalid. The code separately checks the actual response fold, including moving recoil endpoints, as a unit-column kernel validation. No Gaussian replacement is introduced.

3 A bound independent of boosts, geometry acceptance and fine tuning

The retained G2-X scalar-current kernel is

$$\frac{d\sigma_A}{dE_R} = \frac{\lambda_X^2 f_N^2 m_N^2 A^2 m_A}{8\pi p^2 (m_h^2 + 2m_A E_R)^2} F_A^2(\sqrt{2m_A E_R}) \mathbf{1}_{0 < E_R < E_{\text{max}}}, \quad (8)$$

$$E_{\text{max}} = \frac{2m_A p^2}{s}, \quad s = \mu_j^2 + m_A^2 + 2m_A \sqrt{\mu_j^2 + p^2}. \quad (9)$$

Here $f_N = .308$, $m_N = .939$ GeV; natural isotope fractions and the Helm conventions are unchanged from G2-X. Since $F_A^2 \leq 1$, $\mathcal{R} \leq 1$, the propagator denominator is at least m_h^4 , and

$s \geq (\mu_j + m_A)^2$, integration over the *admitted window* gives

$$\sigma_{A,\text{acc}} \leq \frac{\lambda_X^2 f_N^2 m_N^2 A^2 m_A}{8\pi p^2 m_h^4} E_{\text{max}} \quad (10)$$

$$\leq \frac{\lambda_X^2 f_N^2 m_N^2 A^2 m_A^2}{4\pi m_h^4 (\mu_j + m_A)^2} \equiv U_A(\mu_j, \lambda_X). \quad (11)$$

Using E_{max} in this inequality extends a mathematical *constant majorant*, not a nuclear model or calibration outside $[E_0, E_1]$. In particular it does not claim a total coherent cross section at MeV/GeV recoils. The $p = 0$ accepted count is zero because $E_0 > 0$.

With uniform xenon density inferred from the rounded active volume,

$$\ell_{\text{max}} = \sqrt{(2r)^2 + h^2} = 136.333 \text{ cm}, \quad \mathcal{C}_{\text{max}} = \frac{M}{\pi r^2 h} \ell_{\text{max}} \frac{N_A}{M} = 1.78874 \cdot 10^{24} \text{ nuclei cm}^{-2}. \quad (12)$$

Any single straight traversal has at most this column. Thus even if every produced particle is assigned the maximum chord,

$$N_{\text{acc}} \leq 2N_h \mathcal{C}_{\text{max}} \sum_j B_j \sum_A \xi_A U_A(\mu_j, \lambda_X). \quad (13)$$

This is an upper bound for the retained elastic recoil component, not a simulated yield or a background likelihood. The inferred opacity is tiny, consistent with the single-scattering approximation.

For a uniform bound across *all* scales, any on-shell source has $\Lambda < m_h/10$ and $\mu_j \geq 5\Lambda$. The retained vacuum rule gives $\lambda_X^2 \leq 2500\Lambda^4/v^4$. Every term is bounded using

$$\frac{d}{d\Lambda} \log \frac{\Lambda^4}{(5\Lambda + m_A)^2} = \frac{4}{\Lambda} - \frac{10}{5\Lambda + m_A} > 0. \quad (14)$$

Taking the upper endpoint yields

$$\lambda_{\text{sup}} = \frac{m_h^2}{2v^2} = .1292634, \quad (15)$$

$$U_{\text{sup}} = \sum_A \xi_A U_A(m_h/2, \lambda_{\text{sup}}) = 1.33848 \cdot 10^{-36} \text{ cm}^2. \quad (16)$$

$$\boxed{N_{\text{acc}} \leq 2N_h B_{\text{inv}}^{\text{max}} \mathcal{C}_{\text{max}} U_{\text{sup}} = 3.95969 \cdot 10^{-6}.} \quad (17)$$

At $\Lambda \geq m_h/10$ the on-shell source is closed. Hence the bound covers the full 1–30 GeV domain. This envelope does not require its separate factors to be simultaneously attainable: exactly at the threshold the actual production is zero. No search for a specially tuned threshold point is needed.

An explicitly generous stress card, $\sigma_h = 60 \text{ pb}$, $f_N = .326$ and $\mathcal{C}_{\text{max}} = 2 \cdot 10^{24} \text{ cm}^{-2}$, gives $5.35247 \cdot 10^{-6}$. Even replacing $B_{\text{inv}}^{\text{max}}$ by one gives only $3.70064 \cdot 10^{-5}$ under the other nominal premises. These are stress envelopes, not confidence intervals. At least $7.58 \cdot 10^5$ times the nominal source exposure would be needed for three events even under (17)'s impossible perfect interception; no extrapolated accelerator performance is claimed.

At fixed masses, $\Gamma_j = \lambda_X^2 u_j$ and $\sigma_{\text{acc}} = \lambda_X^2 s_j$, so with $x = \lambda_X^2$ a source-linked yield obeys

$$N_{\text{acc}} = \frac{A\lambda_X^4}{\Gamma_{\text{SM}} + U\lambda_X^2}, \quad A, U \geq 0, \quad \frac{d}{dx} \frac{x^2}{\Gamma_{\text{SM}} + Ux} = \frac{x(2\Gamma_{\text{SM}} + Ux)}{(\Gamma_{\text{SM}} + Ux)^2} \geq 0. \quad (18)$$

Thus choosing the allowed coupling ceiling is already optimistic; the old independent-flux feasibility map must not be optimized by choosing an unrelated large source luminosity.

4 Mode information changes with the source

For direct tree-level Higgs production, $\mu_j = \mu_{-j}$, $\Gamma_j = \Gamma_{-j}$, and both scalar decays are isotropic. For any common Higgs momentum distribution, the conditional laboratory four-momentum measures in (5) are identical for j and $-j$. The uniform, diagonal probe has identical matrix elements too. For any measurement z constructed from four-dimensional momentum, direction, flight time and the same scattering response,

$$P(z \mid j = +1) = P(z \mid j = -1), \quad P(j = \pm 1) = \frac{1}{2}, \quad I(j; z) = 0, \quad e_{\text{Bayes}} = \frac{1}{2}. \quad (19)$$

For example $t = d/(c\beta)$, $\beta = p/\sqrt{p^2 + \mu_j^2}$, has the same distribution for both signs. An independent clock or better momentum instrument cannot distinguish equal likelihoods. This is a statement about this production/probe channel, not an assertion of an exact symmetry of the full Φ -holonomy theory at all loops. Different $|j|$ can have different speeds and thresholds; they are not the same problem as resolving the sign of a degenerate pair. The old asymmetric collision sample's 14.057% S2 error is therefore not transferred to this new source.

The selected $|j| = 1$ source closes already at $\Lambda = m_h/(2\sqrt{26}) = 12.2651$ GeV, before the last $j = 0$ threshold. This exposes a misleading intuition from a coupling-only map: the high-scale region with less restrictive on-shell Higgs width also lacks this on-shell source.

5 Decision, next choices, and boundaries

This stage rejects the Run-2 on-shell-Higgs plus single-pass xenon *retained-window elastic* chain as a useful event-count realization. The verdict does not depend on a fitted Gaussian, a chosen boost distribution or an achievable dark-particle focusing system. The verifier checks source normalization, every open signed mode, recoil-bin clipping using the unchanged public response, the continuous bound, and Lorentz/sign identities. Its checks validate implementation, not empirical existence of X .

Do not now build a detailed transport simulation for this already insufficient chain. Two distinct next questions are more productive:

1. Use a collider production/missing-momentum channel to avoid a second tiny scattering probability. On-shell Higgs invisibility constrains an inclusive width, not a resolved tower or its signs. Off-shell thresholds or a tagged missing invariant mass could test masses, but need a separately specified source and analysis.
2. If signed-mode conversion is the indispensable claim, supply a genuinely mode-sensitive preparation or coupling. The original Φ collision correlations could provide it only after a physical Φ source is built; changing the detector resolution does not.

Off-shell Higgs production, astrophysical populations, other targets, electronic/inelastic channels, energies outside the admitted response, trapping/recirculation and additional interactions are not ruled out by the on-shell budget. None is silently added. The GeV-scale classical pump remains unimplemented, and Route F remains independent.

References

- [1] ATLAS Collaboration, JHEP 05 (2023) 028, arXiv:2207.08615v2, Secs. 1–2 and 5.
- [2] ATLAS Collaboration, Phys. Lett. B 842 (2023) 137963, arXiv:2301.10731v2.
- [3] XENON Collaboration, Phys. Rev. Lett. 123 (2019) 251801, arXiv:1907.11485v2; official response release, pinned locally in the G2-X provenance card.